ENTHUSE, LEADER & ACHIEVER COURSE

PHASE : ALL ENTHUSE, LEADER & ACHIEVER SRG MAJOR COURSE

TARGET : PRE-MEDICAL 2026

Test Type : **SRG-MAJOR** Test Pattern : **NEET (UG)**

TEST DATE : 25-05-2026

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	1	3	4	2	4	2	2	2	3	2	2	2	2	2	4	3	3	4	1	4	4	3	3	2	4	1	2	1	4	1
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	3	3	2	1	1	2	1	2	2	3	2	2	3	1	4	3	4	4	2	3	4	1	4	4	3	1	3	4	1	1
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	4	4	3	3	2	1	4	3	2	4	4	1	2	4	4	2	4	2	3	3	3	1	2	2	2	1	2	2	2	4
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	1	2	4	3	1	2	1	3	1	2	4	3	3	3	1	3	1	4	3	4	3	4	3	2	2	3	4	2	4	4
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	1	3	2	3	2	2	3	2	1	3	4	4	3	4	3	3	1	3	4	2	3	2	4	4	2	2	2	4	1	4
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	1	2	4	3	3	3	1	3	4	1	4	1	2	4	3	2	4	3	3	2	4	4	2	3	1	2	4	1	3	3

HINT – SHEET

1. Ans (1)

$n = c / v \Rightarrow v \propto 1/n$

$${}_g\mu_w = \frac{\mu_w}{\mu_g} = \frac{v_g}{v_w}$$

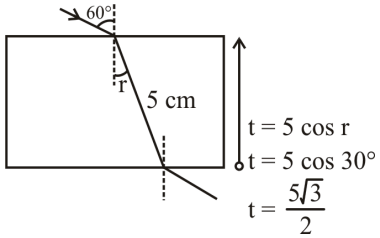
$$\frac{1.33}{1.5} = \frac{v_g}{2.25 \times 10^8}$$

$v_g = 1.95 \times 10^8$ m/s

2. Ans (3)

$\mu_{glass} = \sqrt{3}$, $i = 60°$

$1 \sin 60° = \sqrt{3} \sin r$; $r = 30°$



$$d = \frac{t \sin(i - r)}{\cos r} \Rightarrow d = \frac{5\sqrt{3}}{2} \frac{\sin 30°}{\cos 30°}$$

$$d = \frac{5\sqrt{3}}{2} \times \frac{1}{\sqrt{3}} \Rightarrow d = \frac{5}{2} \text{cm}$$

3. Ans (4)

$$\theta_c = \sin^{-1}\left(\frac{1}{\mu}\right), \quad \mu \propto \frac{1}{\lambda}$$

Violet has smallest wavelength in VIBGYOR $\longrightarrow$

highest $\mu \longrightarrow$ smallest critical angle.

Reason correctly states formula and relation $\mu \propto \frac{1}{\lambda}$.

Reason directly explains why violet has minimum θ_c.

4. Ans (2)

$\delta = 30°$

$A = 60°$ $\quad \delta = i + e - A$

$i + e = 90°$

if $i = e = 45°$, then $\delta_m = 30°$

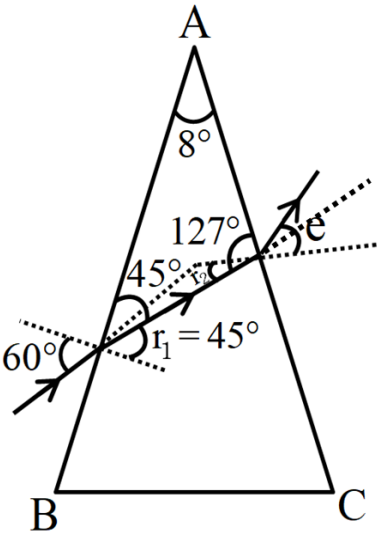
$$\therefore \mu = \frac{\sin\left(\frac{A+\delta_{min}}{2}\right)}{\sin A/2} = \frac{\sin 45°}{\sin 30°}$$

$$= \frac{1/\sqrt{2}}{1/2} = \sqrt{2}$$

If $\delta_{min} < 30° \Rightarrow \mu < \sqrt{2}$

5. Ans (4)

1st surface $\longrightarrow 1 \times \frac{\sqrt{3}}{2} = \sqrt{\frac{3}{2}} \sin r_1 \Rightarrow r_1 = 45°$



$r_2 = 127° - 90° = 37°$

2nd surface

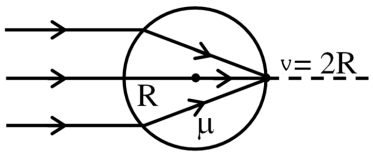
$\sqrt{\frac{3}{2}} \sin 37° = 1 \times \sin e$

$e = \sin^{-1}\sqrt{\frac{27}{50}}$

6. Ans (2)

Formula :

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$



$$\frac{\mu}{2R} - \frac{1}{\infty} = \frac{\mu - 1}{R}$$

$$\frac{\mu}{2R} = \frac{\mu - 1}{R}$$

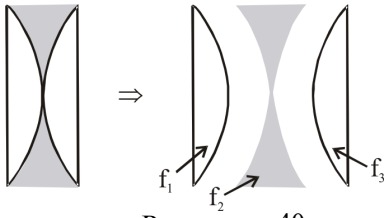
$\mu R = 2\mu R - 2R$

$2R = 2\mu R - \mu R$

$2R = \mu R$

$\mu = 2$

7. Ans (2)



$$f_1 = f_3 = \frac{R}{(\mu - 1)} = \frac{40}{(1.5 - 1)} = \frac{40}{0.5} = 80\,cm$$

$$f_2 = -\frac{R}{2(\mu - 1)} = -\frac{40}{2\left(\frac{4}{3} - 1\right)} = \frac{-20}{\left(\frac{1}{3}\right)} = -60cm$$

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2} + \frac{1}{f_3}$$

$$\frac{1}{f} = \frac{1}{80} - \frac{1}{60} + \frac{1}{80} = \frac{1}{40} - \frac{1}{60} = \frac{3-2}{120}$$

$f = 120$ cm

8. Ans (2)

$f_0 = 1.2$ cm, $f_e = 3$ cm

$u_0 = -1.25$ cm

$$\frac{1}{v_0} - \frac{1}{-1.25} = \frac{1}{1.2} \Rightarrow \frac{1}{v_0} = \frac{1}{1.2} - \frac{1}{1.25}$$

$v_0 = 30$ cm

$$m_0 = \frac{v_0}{u_0} = \frac{30}{-1.25} = -24$$

final image is formed at infinity

So, $m_e = \frac{D}{f_e} = \frac{25}{3}$

magnifying power of microscope,

$M = m_0 m_e = -24 \times \frac{25}{3} = -200$

9. Ans (3)

$I = \frac{I_0}{2}\cos^2 60° = \frac{I_0}{8}$

Percentage of transmitted intensity:

$\frac{I}{I_0} \times 100 = \frac{1}{8} \times 100 = 12.5\%$

10. Ans (2)

$d \sin\theta = \lambda =$ constant

As d decreases, width increases

$\Rightarrow$ intensity decreases

11. Ans (2)

In Young's experiments fringe width for dark and bright fringes are same while in Young's double slit experiment when a white light as a source is used, the central fringe is white around which few coloured fringes are observed on either side.

12. Ans (2)

From I polariod $\Rightarrow \frac{I_0}{2}$

From II $\Rightarrow \frac{I_0}{2}\cos^2 30^o$

$= \frac{I_0}{2} \times \frac{3}{4} = \frac{3I_0}{8}$

From III $\Rightarrow \frac{3I_0}{8}\cos^2 30^o$

$= \frac{3I_0}{8} \times \frac{3}{4} = \frac{9I_0}{32}$

From IV $\Rightarrow \frac{9I_0}{32}\cos^2 30^o$

$= \frac{9I_0}{32} \times \frac{3}{4} = \frac{27I_0}{128}$

13. Ans (2)

No. of photoelectrons $= \frac{1}{100} \times \frac{IA\lambda}{hc}$

$= \frac{1}{100} \times \frac{1 \times 10^{-4} \times 3 \times 10^{-7}}{2 \times 10^{-25}}$

$\approx 1.5 \times 10^{12}$ per sec.

14. Ans (2)

$\lambda = \frac{h}{\sqrt{2mE}} \Rightarrow \lambda \propto \frac{1}{\sqrt{K}}$

Let initial kinetic energy = K, initial wavelength = λ

After increase :

K' = 16 K

New wavelength :

$\lambda' = \frac{\lambda}{\sqrt{16}} = \frac{\lambda}{4}$

Decrease in wavelength :

$\Delta\lambda = \lambda - \lambda' = \lambda - \frac{\lambda}{4} = \frac{3\lambda}{4}$

Percentage change :

$\frac{\Delta\lambda}{\lambda} \times 100 = \frac{3}{4} \times 100 = 75\%$

(The wavelength decreases by 75%)

15. Ans (4)

$13.6 + 2.4 = \frac{hc}{\lambda}$

16. Ans (3)

Since $R \propto A^{1/3} \Rightarrow R^3 \propto A$ and

Volume $\propto R^3$ or Volume $\propto A$

Hence density $= \frac{\text{mass}}{\text{volume}} = \frac{A \times m}{\frac{4}{3}\pi R^3} = \frac{A \times m}{\frac{4}{3}\pi R_0{}^3 A}$

$= \frac{3m}{4\pi R_0{}^3} = 2.3 \times 10^{17} \text{kg/m}^3$ (constant)

it shows nuclear density is independent of A.

Therefore nuclear density is nearly same for all nuclei.

So, statement A is correct But in statement B:

$R^2 \propto A^{1/6}$, which is incorrect.

17. Ans (3)

$Ne^{20} \longrightarrow 2He^4 + C^{12}$

$8.03 \times 20 \longrightarrow 2 \times 7.07 \times 4 + 7.68 \times 12$

$Q_{value} = (BE)_{product} - (BE)_{react} = -11.88$ MeV

So 11.88 MeV will have to be supplied.

18. Ans (4)

Let's assume that there are 'a' number of 'α' particle and 'b' number of 'β^-' particles emitted in the given nuclear reaction:

$_{93}Np^{237} \longrightarrow {}_{90}Th^{229} + a({}_2He^4) + b\,({}_{-1}e^0)$

On balancing the atomic number

$\Rightarrow 93 = 90 + a(2) + b(-1)$

$\Rightarrow 93 = 90 + 2a - b$

$\Rightarrow 2a - b = 3$ .. (i)

On balancing the mass number

$\Rightarrow 237 = 229 + a(4) + b(0)$

$\Rightarrow 237 = 229 + 4a + 0$

$\Rightarrow 4a = 8 \Rightarrow a = 2$

Put this value of 'a' into equation (i), to get -

$2(2) - b = 3$

$\Rightarrow 4 - b = 3 \Rightarrow b = 1$

So, a = 2 & b = 1

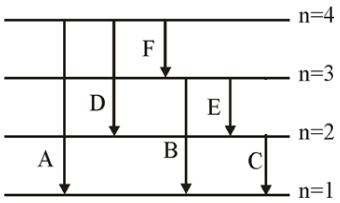
19. Ans (1)

When electron beam of 12.75 eV is incident over H-gaseous.

Energy of H-atoms will be

$= -13.6 + 12.75 = -0.85$ eV

The corresponds to n = 4 excited state



Clearly, emitted radiation lies in lyman series (A, B and C) few emissions are is Balmer series (D and E). One of them (n=4 to n=3) lies in paschen series (F).

20. Ans (4)

The P-N junction will conduct only when it is forward biased i.e. when $-$ 5V is fed to it, so it will conduct only for 3rd quarter part of signal shown and when it conducts potential drop 5 volt will be across both the resistors, so output voltage across R_2 is 2.5V.

$\therefore V_0 = -2.5$V

21. Ans (4)

Maximum wavelength detectable, $\lambda_{max} = \dfrac{hc}{E_g}$

$\lambda_{max} = \dfrac{12400}{2.8}$

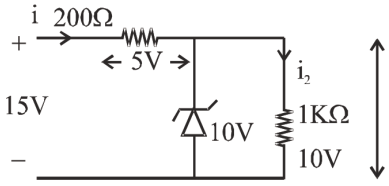
$\lambda_{max} \approx 4428.5$ Å

For detection, the incident wavelength must be less than or equal to λ_{max}.

Among the options, only 4000 Å is less than 4428.5 Å.

22. Ans (3)

$v_{in} > v_z$,hence diode may be at breakdown

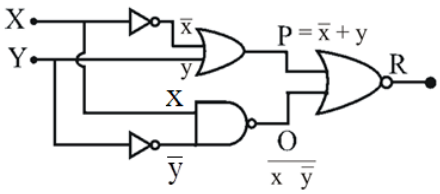


$i = 5/200 = 0.025A = 25$ mA

$i_L = 10/10^3 = 10$mA

$i_Z = i - i_L = 25 - 10 = 15$mA

23. Ans (3)



$R = \overline{(\bar{x}+y)+\overline{x\bar{y}}} = \overline{(\bar{x}+y)+(\bar{x}+\bar{\bar{y}})}$

$R = \overline{\bar{x}+y} = \bar{\bar{x}}.\bar{y} = x.\bar{y}$

For R = 1 ; x = 1, y = 0

24. Ans (2)

$B = \mu H \Rightarrow 8 \times 10^{-3} \times 160 = 1.28$ Wb/m^2

25. Ans (4)

$W = MB(1 - \cos 60°) = \dfrac{MB}{2} = 3$

$\Rightarrow MB = 6$

$\tau = MB \sin 60° = 6 \times \dfrac{\sqrt{3}}{2} = 3\sqrt{3}$Nm

26. Ans (1)

Side of square: a = 2m

Area: $A = a^2 = 4m^2$

Current: $I = \dfrac{1}{2}$A

Magnitude of magnetic moment:

$$M = IA = \frac{1}{2} \times 4 = 2A\ m^2$$

Direction of area vector is:

$$\hat{n} = \frac{1}{2}\hat{i} - \frac{\sqrt{3}}{2}\hat{k}$$

Thus:

$$\vec{M} = 2\left(\frac{1}{2}\hat{i} - \frac{\sqrt{3}}{2}\hat{k}\right)$$

$\vec{M} = \hat{i} - \sqrt{3}\,\hat{k}$ Am^2

27. Ans (2)

Electric field between charged plates: $E = \sigma / \varepsilon_0$

Electric force: $F_E = eE$

Magnetic force: $F_B = e\ v\ B$

Condition for straight motion: $F_E = F_B$

Calculation:-

For straight line motion, electric and magnetic forces must balance,

$eE = e\ v\ B$

Substitute $E = \sigma / \varepsilon_0$,

$e(\sigma / \varepsilon_0) = e\ v\ B$

$\Rightarrow v = \sigma / (\varepsilon_0 B)$

Time taken to cross length ℓ,

$t = \dfrac{\text{Distance}}{\text{Speed}} = \dfrac{\ell}{v}$

$\Rightarrow t = \ell / [\sigma / (\varepsilon_0 B)]$

$\Rightarrow t = (\varepsilon_0\ \ell\ B) / \sigma$

28. Ans (1)

$\vec{F} = q\left(\vec{v} \times \vec{B}\right)$

The force acting on the electron is given by:

$\vec{F} = -e(2\hat{i} + 3\hat{j}) \times 4\hat{j}$

$= -e[2(4\hat{k}) + 3(0)]$

$= -e\hat{k}$

- Since force is perpendicular to velocity, only the direction of velocity changes, not the speed
- The motion is helical as the velocity has components parallel and perpendicular to $\vec{B}$.

Final Answer

(a) velocity will change

(b) speed does not change

(d) path must be helix

29. Ans (4)

$B = \dfrac{\mu NI}{2\pi R}$

$1 = \dfrac{\mu_r \times 4\pi \times 10^{-7} \times 400 \times 2}{40 \times 10^{-2}}$

$\Rightarrow 1 = 4\pi \times 10^{-4} \times 2\ \mu_r$

$\Rightarrow \mu_r = \dfrac{10^4}{8\pi} = 398 \simeq 400$

30. Ans (1)

As we know the flux through a coil

$\phi_{total} = NBA\cos\theta$ -------- (1)

where, θ is the angle between $\vec{B}$ & $\vec{A}$

If the coil is rotating about its diameter perpendicular to the magnetic field,

The induced emf will be,

$e = \dfrac{-d\,\phi_{total}}{dt}$

$= -NBA\ \dfrac{d\cos\theta}{dt}$

$e = -NBA[-\sin\theta]\ \dfrac{d\theta}{dt}$

$e = +NBA\omega\sin\omega t$

$\because \dfrac{d\theta}{dt} = \omega = \text{constant}$, So, $\theta = \omega t$

And from equation --- (1)

$\phi_{total} = NBA\cos\omega t$

Phase difference between flux and induce EMF is 90 degree which implies when flux is maximum, EMF is zero and when EMF is maximum, flux is zero.

31. Ans (3)

When a rod rotates in a uniform magnetic field, an EMF is induced across its length. The potential difference between the pivot (center O) and any point at a distance l is given by $\dfrac{1}{2}B\omega\ell^2$.

To find the potential difference between two specific points P and Q, we calculate the potential of each relative to the center O and then find the difference between them.

Calculation:

1. Potential at point P (distance L/5from O):

$V_O - V_P = \dfrac{1}{2}B\omega\left(\dfrac{L}{5}\right)^2 = \dfrac{1}{2}B\omega\dfrac{L^2}{25} = \dfrac{B\omega L^2}{50}$

2. Potential at point Q (distance 4L/5 from O):

$V_O - V_Q = \dfrac{1}{2}B\omega\left(\dfrac{4L}{5}\right)^2 = \dfrac{1}{2}B\omega\dfrac{16L^2}{25} = \dfrac{16B\omega L^2}{50}$

3. Potential difference between P and Q:

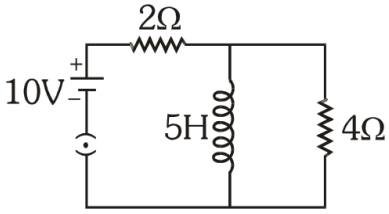
$|V_P - V_Q| = (V_O - V_Q) - (V_O - V_P)$

$\Delta V = \dfrac{16B\omega L^2}{50} - \dfrac{B\omega L^2}{50} = \dfrac{15B\omega L^2}{50}$.

4. Simplify:

$\Delta V = \dfrac{3}{10}BL^2\omega$

32. Ans (3)



Just after closing the key, the inductor opposes sudden change in current and behaves as an open circuit.

Hence, current flows only through the 2 Ω and 4 Ω resistors in series.

Equivalent resistance = 2 + 4 = 6 Ω.

Current through 4 Ω resistance = $\frac{10}{6}A = \frac{5}{3}A$.

33. Ans (2)

$B = \mu_0 nI$

$\phi = N\,\mu_0 nI\,A$

$emf = N\,\mu_0 n\,A\,\frac{\Delta I}{\Delta t} = 9\text{ mV}$

Here $N = 300$, $n = 10^3$

$A = 1.2 \times 10^{-3}\text{ m}^2$, $\Delta I = 4A$

$\Delta t = 0.2$ sec

34. Ans (1)

quality factor:

$Q = \frac{\omega_0 L}{R}$

Also,

$Q = \frac{\omega_0}{\Delta\omega}$

So,

$\Delta\omega = \frac{R}{L}$

But

$\Delta\omega = \omega_2 - \omega_1 = 2\pi(f_2 - f_1)$

Thus,

$f_2 - f_1 = \frac{R}{2\pi L}$

35. Ans (1)

$R = 500\ \Omega$

$X_L = \omega \times L = 1000 \times 0.5 = 500$

$\cos\phi = \frac{R}{\sqrt{R^2 + X_L^2}}$

36. Ans (2)

$$I_{rms}^2 = \frac{\int_2^4 I^2 dt}{\int_2^4 dt} = \frac{\int_2^4 4t dt}{\int_2^4 dt} = \frac{\left(\frac{4t^2}{2}\right)_2^4}{(2)} = \frac{2[4^2 - 2^2]}{2}$$

$I_{rms}^2 = 12$, $I_{rms} = 2\sqrt{3}A$

37. Ans (1)

$I = \frac{1}{2}\varepsilon v E_0^2$

$v = \frac{1}{\sqrt{\mu\varepsilon}}$

Given:

$E_0 = 50$, $\mu = 4\mu_0$, $\varepsilon = \varepsilon_0$

$v = \frac{1}{\sqrt{4\mu_0\varepsilon_0}} = \frac{1}{2\sqrt{\mu_0\varepsilon_0}} = \frac{c}{2}$

$I = \frac{1}{2}\varepsilon_0 \cdot \frac{c}{2} \cdot (50)^2$

$= \frac{1}{4}\varepsilon_0 c \cdot 2500$

$= \frac{2500}{4} \times \varepsilon_0 c = 625 \times (8.85 \times 10^{-12} \times 3 \times 10^8)$

$= 625 \times 2.655 \times 10^{-3} \approx 1.65\text{ W/m}^2$

38. Ans (2)

For the question, infrared is the least energetic thus having biggest wavelength (λ) & gamma rays are most energetic thus having smallest wavelength (λ). Ultraviolet is lesser energetic than X-rays, so UV has larger wavelength than X-ray.

39. Ans (2)

$$\frac{\mu_3}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R_1} + \frac{\mu_3 - \mu_2}{R_2}$$

OR

For AiR – oil surface

$u = -24cm$; $u_1 = 1$, $u_2 = \frac{7}{4}$, $R = +6cm$

$$\frac{u_2}{v} - \frac{u_1}{u} = \frac{u_2 - u_1}{R}$$
$$\frac{7}{4v} - \frac{1}{-24} = \frac{\frac{7}{4} - 1}{6}$$
$$\frac{7}{4v} + \frac{1}{24} = \frac{3}{4 \times 6}$$
$$\frac{7}{4v} = \frac{1}{8} - \frac{1}{24} = \frac{3-1}{24}$$
$$\frac{7}{4v} = \frac{1}{12} \Rightarrow \boxed{v = +21cm}$$

This v image act as object for oil - water surface

$u_1 = \frac{7}{4}$ $u_2 = \frac{4}{3}$ $R = \infty$

$u = 21$ cm

$$\frac{u_2}{v} - \frac{u_1}{u} = \frac{u_2 - u_1}{R}$$
$$\frac{4}{3v} - \frac{7}{4(21)} = \frac{u_2 - u_1}{\infty}$$
$$\frac{4}{3v} - \frac{1}{12} = 0$$
$$\frac{4}{3v} = \frac{1}{12}$$

$\boxed{v = +16cm} \Rightarrow$ from water surface

from Bottom of tank

$\Rightarrow x = 18 - 16 = 2cm$

40. Ans (3)

Inequilibrium

$$\text{Tension} = T = \sqrt{(mg)^2 + \left(\frac{P}{C}\right)^2} = ma$$

$$t = 2\pi\sqrt{\frac{\ell}{T/m}} = 2\pi\frac{\sqrt{\ell}}{\left(g^2 + \left(\frac{P}{mC}\right)^2\right)^{1/4}}$$

41. Ans (2)

As $K.E_{max}$ and stopping potential is lineraly dependent on frequency of photon, so, when frequency is doubled the $K.E_{max}$ and stopping potential becomes more than double.

42. Ans (2)

$r = 2R$, $5\mu T = \frac{\mu_0 I}{2\pi(2R)} \Rightarrow \frac{\mu_0 I}{\pi R} = 20\mu T$

At distance $\frac{R}{3}$ from suface means $\frac{2R}{3}$ from centre.

$$r = \frac{R}{3}, B = \frac{\mu_0 I r}{2\pi R^2} = \frac{\mu_0 I\left(\frac{2R}{3}\right)}{2\pi R^2} = \frac{\mu_0 I}{3\pi R}$$

$$= \frac{1}{3} \times 20\mu T = \frac{20}{3}\mu T$$

43. Ans (3)

Question Explanation: Motion of a charge in combined electric and magnetic fields.

Concept: Lorentz Force and Kinematics.

Solution: The electric field $\vec{E}$ accelerates the particle in the x-direction, while the magnetic field $\vec{B}$ (parallel to $\vec{E}$) does not exert force on the initial velocity component in the y-direction. The total speed is the vector sum of the constant y-velocity and the increasing x-velocity.

Calculation: Acceleration $a_x = \frac{qE_0}{m}$. Velocity $v_x = a_x t = \frac{qE_0 t}{m}$.

- Constant velocity $v_y = v_0$.
- Speed $v = \sqrt{v_x^2 + v_y^2}$. We need $v = 2v_0$.
- $4v_0^2 = v_x^2 + v_0^2 \Rightarrow v_x^2 = 3v_0^2 \Rightarrow v_x = \sqrt{3}v_0$.
- $\frac{qE_0 t}{m} = \sqrt{3}v_0 \Rightarrow t = \frac{\sqrt{3}mv_0}{qE_0}$.

Final Answer: Option (3)

44. Ans (1)

$$\phi = \int_0^{R/2} B\,(2\pi r)dr = B_0 t 2\pi \left(\frac{r^3}{3}\right)_0^{R/2}$$

$$= B_0\,(2\pi)\,\frac{1}{3}\left(\frac{R^3}{8}\right)t$$

$$\because E\left(2\pi\left(\frac{R}{2}\right)\right) = \frac{-d\phi}{dt} = -B_0\,(2\pi)\,\frac{R^3}{24}$$

$$\Rightarrow E = \frac{B_0R^2}{12}$$

45. Ans (4)

From photo electric equation,

$$KE_{max} = h\nu - \phi = \frac{12400}{4000} - 1.9 = 1.2 \text{ eV}$$

Now from energy conservation,

$e + \alpha = He^+ + \text{Photon}$ .....(i)

$\because$ Energy of He^+ ion in 4th excited state

$$E = -13.6 \text{ eV}\left(\frac{Z^2}{n^2}\right)$$

$$= -13.6\,\frac{(2)^2}{(5)^2} \text{ eV} = -2.176 \text{ eV}$$

Substituting values in eq(1)

$1.2 + 0 = -2.176 + E_{Photon}$

$\Rightarrow E_{photon} = 1.2 + 2.176 = 3.376$ eV

46. Ans (3)

Rs. (5) (c) is the correct answer

Mohr salt $\longrightarrow FeSO_4.(NH_4)_2SO_4.6H_2O$

Carnallite $\longrightarrow KCl.MgCl_2.6H_2O$

Heteroleptic complex $\longrightarrow [Fe(H_2O)_5NO]SO_4$

Potash alum $\longrightarrow K_2SO_4.Al_2(SO_4)_3.24H_2O$

47. Ans (4)

Concept based on synergic bonding.

The stronger the metal-carbon bond, the weaker will be C–O bond.

48. Ans (4)

$A \longrightarrow [Co(H_2O)_6]Cl_3$

$B \longrightarrow [Co(H_2O)_5Cl].Cl_2.H_2O$

$C \longrightarrow [Co(H_2O)_4Cl_2]Cl.\ 2H_2O$

49. Ans (2)

Explanation:

The question asks you to identify the incorrect stability order among the given sets of coordination complexes.

Concept: (Stability chelate effect, ligand strength, Metal ion size, Metal oxidation state of coordination compounds)

- **Chelate effect:** Polydentate ligands (en, trien) create more stable complexes. (Correct)
- **Ligand strength:** Stronger field ligands (NO_2^-) create more stable complexes than weaker ones (NH_3, H_2O). (Incorrect)
- **Metal ion size:** Stability increases down a group for similar complexes.
- **Metal oxidation state:** Higher oxidation states generally increase complex stability.

Final Answer: Option (2)

50. Ans (3)

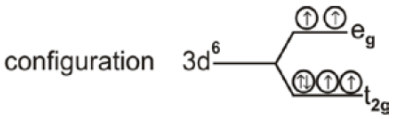
Explaining the question : Identify which of the listed cobalt complexes are chiral and thus exhibit optical isomerism.

Concept : Optical Isomerism

Solution : Cis-$[Co(en)_2Cl_2]^+$ is chiral due to the adjacent arrangement of ligands lacking a plane of symmetry. $[Co(en)_3]^{3+}$ is also chiral as the tris-chelated structure is non-superimposable on its mirror image.

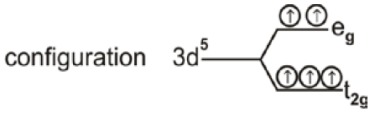
51. Ans (4)

1. $\underset{\text{WFL}}{[\overset{\text{II}}{Fe}(H_2O)_6]^{+2}}$



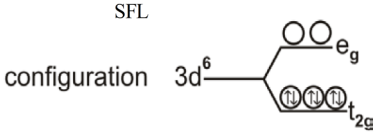
$t_{2g}^4 e_g^2$ (S)

2. $\underset{\text{WFL}}{[\overset{\text{II}}{Mn}(H_2O)_6]^{+2}}$



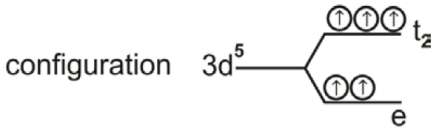
$t_{2g}^3 e_g^2$ (Q)

3. $\underset{\text{SFL}}{[\overset{\text{III}}{Co}(NH_3)_6]^{+3}}$



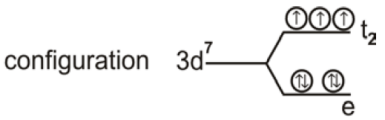
$t_{2g}^6 e_g^0$ (P)

4. $\underset{\text{WFL}}{[\overset{\text{III}}{Fe}Cl_4]^{\ominus}}$



$e^2 t_2^3$ (R)

5. $\underset{\text{WFL}}{[\overset{\text{II}}{Co}Cl_4]^{-2}}$



$e^4 t_2^3$ (None)

52. Ans (1)

Application based

53. Ans (4)

A. CrO_4^{2-} is tetrahedral

B. $2Na_2CrO_4 + 2H^+ \longrightarrow Na_2Cr_2O_7 + 2Na^+ + H_2O$

C. As per NCERT, green manganate is paramagnetic with 1 unpaired electron.

D. Statement is correct

E. Statement is correct

54. Ans (4)

Actionids on reaction with boiling water gives oxide and hydride.

55. Ans (3)

Application based

56. Ans (1)

Cr^{+6} is less stable than Mo^{+6} thats why it is more oxidising.

57. Ans (3)

Due to small size of Cu^{+2} it shows more $\Delta H_{hydration}$.

58. Ans (4)

Highest fluoride of Mn is MnF_4.

59. Ans (1)

A. Question Explanation

Identifying the product formed when $KMnO_4$ reacts with conc. H_2SO_4 and its decomposition.

B. Concept

Concentrated acids convert permanganate into unstable higher oxides like Mn_2O_7.

D. Solution

When $KMnO_4$ is treated with concentrated H_2SO_4, it forms dark green oily Mn_2O_7, an explosive oxide of manganese. Upon heating, Mn_2O_7 decomposes to form MnO_2. Thus, the pair of compounds produced are Mn_2O_7 and MnO2 respectively.

E. Final Answer OPTION 1

60. Ans (1)

CoS is black colour while

$Sb_2S_3 \longrightarrow$ orange

SnS $\longrightarrow$ brown

& CdS $\longrightarrow$ yellow

61. Ans (4)

$CuSO_4 + KI \longrightarrow Cu_2I_2 + I_2 + K_2SO_4$

I_2 + Starch $\longrightarrow$ Blue colour

62. Ans (4)

Question Explanation

Question is asking about correct statement and about their relation

Concept

This concept based on Brown ring test, Test for radicals.

Solution

The common qualitative test for nitrate ions (NO_3^-) is the brown ring test. However, nitrite ions (NO_2^-) also produce a brown ring with the reagents used in this test (ferrous sulfate and concentrated sulfuric acid), and can do so even more readily (sometimes in dilute acid without concentrated H_2SO_4). This interference means that a direct test for NO_3^- in the presence of NO_2^- would give a false positive or an ambiguous result for NO_3^-.

So statement 1 is false.

By observing the color of the aqueous solution after exposing it to the respective substances (e.g., bubbling the gases through water), one can distinguish between Br_2 and NO_2. Bromine will yield a brown (or yellowish-brown) solution, while nitrogen dioxide will yield a colorless solution as it reacts with the water.

so statement 2 is true

Final Answer (4)

63. Ans (3)

Question Explanation: Dilute H_2SO_4 helps identify anions that release characteristic gases.

Concept: Anions giving gases with dilute acids are detected by dil. H_2SO_4.

Solution:

- $CO_3^{2-} + H_2SO_4 \longrightarrow CO_2\uparrow$ (effervescence)
- $SO_3^{2-} + H_2SO_4 \longrightarrow SO_2\uparrow$ (pungent smell)
- $NO_2^- + H_2SO_4 \longrightarrow NO_2\uparrow$ (brown gas)
- NO_3^-, Cl^-, I^- do **not** give characteristic gases with dilute H_2SO_4

Final Answer: Option (3)

64. Ans (3)

Question Explanation :

Question is asking about why the concentration of sulphide is kept low while performing group II radical test.

Concept :

This question is based on Radical test.

Solution :

The concentration of sulphide is kept low while performing group II radical test to achieve selective precipitation and prevent the precipitation of cations from subsequent analytical groups, especially Group IV (and sometimes Group III).

Hence,

Option (3) is correct : Radicals of group IV can also give the test at high concentration of sulphide, so for group II HCl is used to decrease the concentration of S^{2-} ion.

65. Ans (2)

Explaining the Question :-

Question is asking about matching of compounds

Concept:-

Colour of precipitate, Colour of solution, general reaction

Solution :-

$ZnCl_2 + H_2S$ (basic) $\longrightarrow ZnS(s) + 2HCl$

ZnS(zinc sulphide) is a **white precipitate** that forms in the reaction.

$Cu^{2+}(aq) + 2OH^-(aq) \longrightarrow Cu(OH)_2(s)$

$Cu(OH)_2(s) + 4NH_3(aq) \longrightarrow [Cu(NH_3)_4]^{+2}(aq) + 2OH^-(aq)$

The **blue precpitate** of $Cu(OH)_2$ **dissolves** and the solution turns **deep blue** due to the **$[Cu(NH_3)_4]^{2+}$** complex.

$FeCl_3 + 3KSCN \longrightarrow Fe(SCN)_3 + 3KCl$

The **ferric thiocyanate complex** is **blood-red** in color.

$Pb(NO_3)_2 + 2KI \longrightarrow PbI_2 + 2KNO_3$

Lead(II) iodide (PbI_2) is the **yellow precipitate**.

Final Answer :- 2

66. Ans (1)

Both Xe & O_2 here act as counter cation Xe^+ & O_2^+

67. Ans (4)

Question Explanation: Select the correct statements regarding Group 16 elements.

Concept: Properties and oxides of the oxygen family.

Solution:

Statement A is correct as Group 16 elements form EO_2 and EO_3 oxides which are generally acidic (Te and Po oxides show amphoteric character but are included in the general trend).

Statement B is correct because SO_2 is a reducing agent (S goes from +4 to +6) while TeO_2 is an oxidizing agent due to the inert pair effect.

Statement C is false; reducing property of hydrides increases down the group (H_2S to H_2Te).

Statement D is false; ozone (O_3) has six lone pairs of electrons.

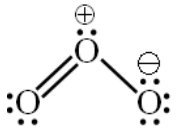
Oxidation state change: S (+4 ⟶ +6) is oxidation; Te (+4 ⟶ 0/+2) is reduction.

2. Ozone lone pairs: Central O (1) + Terminal double bonded O (2) + Terminal single bonded O (3) = 6.

(A) All group 16 elements form oxides of the EO_2 and EO_3 type where E = S, Se, Te or PO.

(B) SO_2 is reducing while TeO_2 is an oxidising agent.

(C) The reducing property increases from H_2S to H_2Te down the group.

(D)  have six lone pairs

Final Answer: Option (4)

68. Ans (3)

TAE : Anti elimination of erythreo will give trans-alkene.

69. Ans (2)

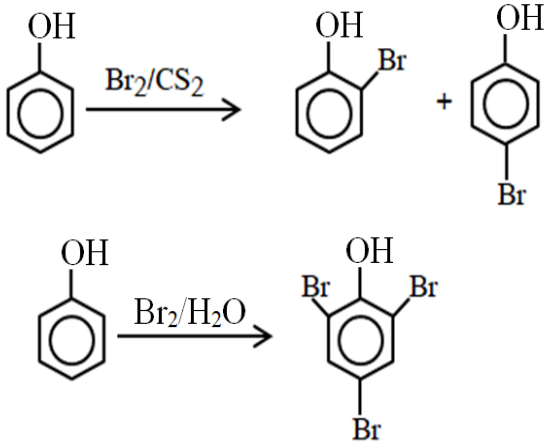
Solution:

$$CH_3-CH_2-Cl+NaI \xrightarrow{\text{Acetone}} CH_3CH_2-I+NaCl$$

NaCl is precipitate in acetone it facilitate forward reaction according to Le-Chatelier principle.

Final answer: option (2)

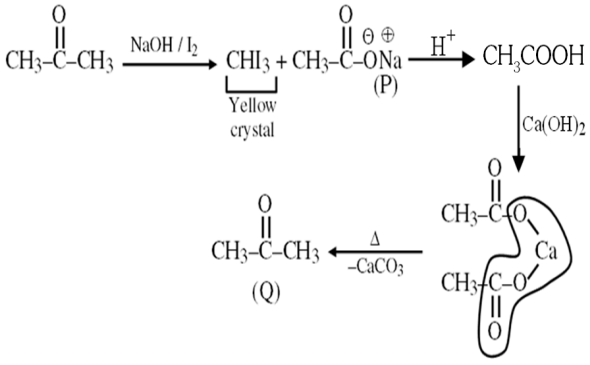
70. Ans (4)



White crystal

71. Ans (4)

Solution :-



Product (P) is CH_3COOH product (Q) is acetone (CH_3COCH_3) yellow crystal of CHI_3 obtain along with (P)

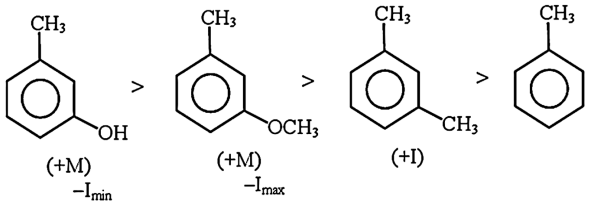
Therefore all of these are correct.

Final Answer :- (4)

72. Ans (1)

Solution :-

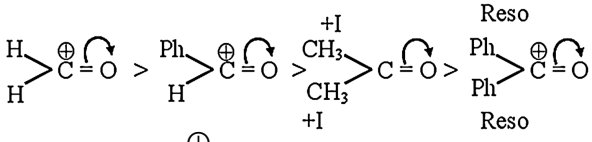
(a)



Rate of ESR $\propto e^-$ density on benzene ring $\propto \frac{EDG}{EWG}$

Given order is correct.

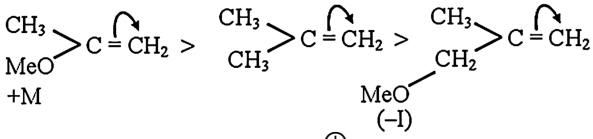
(b)



Rate of NAR $\propto \oplus$ne charge on carbonyl carbon $\propto \frac{1}{\text{Steric hindrance}}$

Given order is correct.

(c)

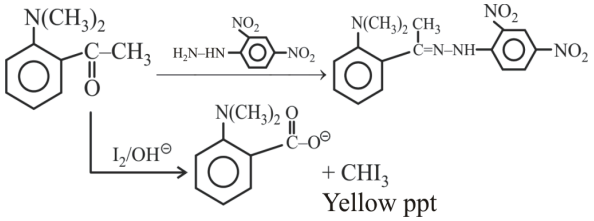


Rate of EAR $\propto$ stability of $C^{\oplus}$

Therefore (a), (b), (c) all are correct.

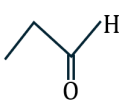
Final Answer :- (1)

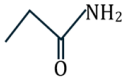
73. Ans (2)



74. Ans (4)

Solution :

(1)  $\xrightarrow[H_2O]{\text{DIBAL-H}}$  ⇒ Correct

(2)  $\xrightarrow{Br_2 / KOH}$ $CH_3-CH_2-NH_2$ ⇒ Correct

(3) $CH_3-CHO \xrightarrow[\Delta]{\text{dil NaOH}} CH_3-CH=CH-CHO$ ⇒ Correct

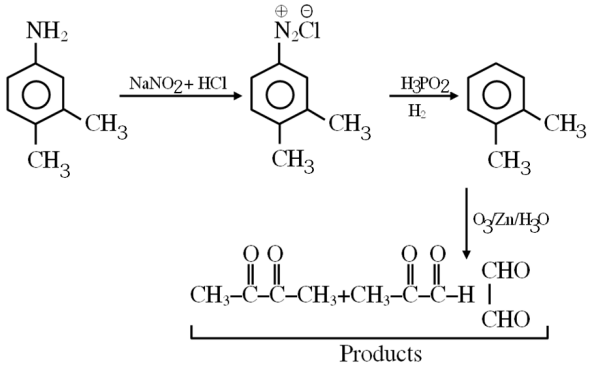
(4)  $\xrightarrow{O_3 / H_2O}$ $2CH_3COOH$ ⇒ Correct

it is oxidative ozonolysis.

Final Answer : option (4)

75. Ans (4)

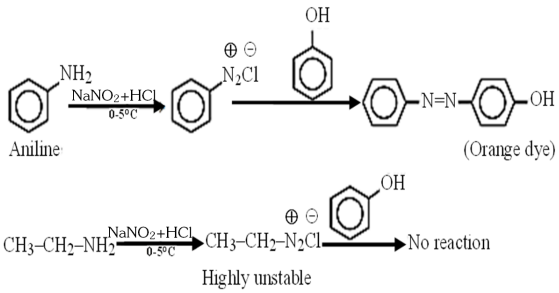
Solution:



Final Answer: (4)

76. Ans (2)

Solution :



Reaction with $NaNO_2$/HCl(0-5°C) followed by phenol clearly distinguishes aniline and ethyl amine.

Final Answer : Option (2)

77. Ans (4)

Mutarotation

The β and α glucose have different specific rotations.

When either is dissolved in water, their rotation changes until the same fixed value results.

This is called mutarotation.

Final answer: option (4)

78. Ans (2)

It is test of amide linkage.

79. **Ans (3)**

Column-I Item	Column-II Match	Reason
(A) Best and latest technique	(i) Chromatography	Modern separation method
(B) Lassaigne's test for N	(iv) Prussian blue coloured complex	NaCN + Fe ⟶ Prussian blue
(C) Lassaigne's test for S	(iii) Violet coloured complex	Na2S + Nitroprusside ⟶ Violet
(D) Aniline purified from water	(v) Steam distillation	Aniline is steam volatile
(E) Glycerol purified from spent lye	(ii) Vacuum distillation	Glycerol decomposes at bp

The Correct matching order is (A)-(i), (B)-(iv), (C)-(iii), (D)-(v), (E)-(ii). This matches Option 3.D)

Final Answer: Option (3)

80. **Ans (3)**

Explanation:

- Na_2O ⟶ not used.
- Na_2SO_4 ⟶ drying agent, not adsorbent. Al_2O_3 (Alumina) ⟶ widely used as an
- adsorbent in column chromatography.
- NaCl ⟶ not used.

Final Answer: Option (3)

81. **Ans (3)**

Solution: Sodium fusion is a powerful Reducing Process; elements are converted into their reduced ionic forms.

- Sodium fusion converts Nitrogen and Sulfur into Cyanide (CN^-), Sulfide (S^{2-}), or Thiocyanate (CNS^-).
- Since molten sodium is a strong Reducing Agent, it cannot form oxidized ions like Nitrite (NO_2^-) or Nitrate (NO_3^-).

The species which is not present in the sodium extract is (NO_3^-)

Final Answer: Option (3)

82. **Ans (1)**

Solution:

Basicity: Oxygen is smaller and more electronegative than Sulfur. The negative charge on RO^- is more concentrated, making it a stronger proton acceptor. Thus, RO^- is more basic than RS^-.

Nucleophilicity: In protic solvents, Sulfur is larger and more polarizable than Oxygen. Its valence electrons are less tightly held, allowing easier overlap with an electrophile's carbon.

Reaction:

$RS^- + R' - X \longrightarrow R - S - R' + X^-$ (Thiolate acts as a superior nucleophile).

Final Answer: Option (1)

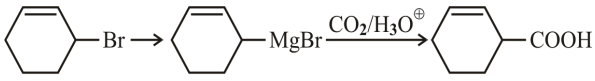
83. **Ans (2)**

Boiling points depend on molecular polarity. The ortho isomer (a) has the highest dipole moment, while the para isomer (c) is symmetric with a zero dipole moment. Thus, the order is **ortho > para > meta**.

Boiling point: (a) > (c) > (b)

Final answer: Option (2)

84. **Ans (2)**



(A)

85. **Ans (2)**

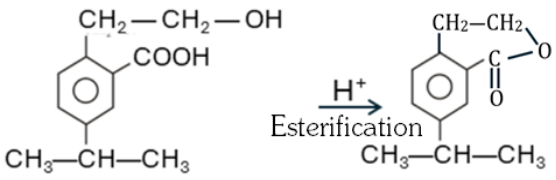
Solution :

Step 1: $NaBH_4$ selectively reduces the aldehyde ($-CH_2CHO$) to a primary alcohol ($-CH_2CH_2OH$), leaving the -COOH and

1. isopropyl groups intact (Product A).

Step 2: Heating with H^+ causes intramolecular esterification between the alcohol and the carboxylic acid, forming a

2. stable six-membered lactone ring.



Final Answer: Option (2)

86. Ans (1)

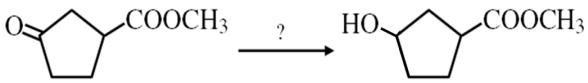
Solution: Ester hydrolysis involves a nucleophilic attack on the carbonyl carbon. The reactivity increases if the leaving group (phenoxide ion) is stabilized by **Electron Withdrawing Groups (EWG)**.

- **I ($-NO_2$):** Strong EWG (−M, −I); most stable phenoxide; highest reactivity.
- **II (−Cl):** EWG (−I > +M); stabilizes phenoxide.
- **III ($-CH_3$):** EDG (+I, hyperconjugation); destabilizes phenoxide.
- **IV ($-OCH_3$):** Strong EDG (+M); least stable phenoxide; lowest reactivity.

Final Answer: Option (1)

87. Ans (2)

Solution :-



This conversion required selective reagent to transformation.

(1) $LiAlH_4$:- This is strong reducing agent that reduces both ketone and ester to alcohol.

So, $LiAlH_4$ is not used for this conversion.

(2) $NaBH_4$:- Mild reducing agent that selectively ester under typical conditions, so it is suitable reagent for above conversion.

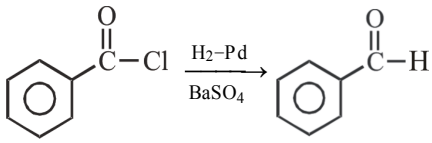
(3) Catalytic reduction :- This reagent can reduce ketones, but it can also reduce ester so it might not be selective for just ketone in presence of ester.

(4) Zn−Hg/HCl :

Therefore $NaBH_4$ is suitable reagent for this conversion.

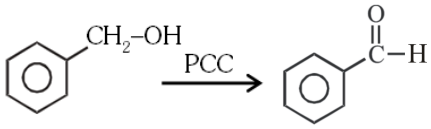
Final Answer :- (2)

88. Ans (2)

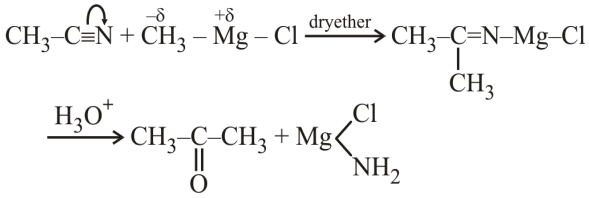
(i) 

Rosenmund Reaction

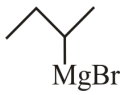
(ii) $CH_3CH_2OH \xrightarrow{PCC} CH_3-CHO$

(iii) 

(iv)



89. Ans (2)

A ⟶ 

B ⟶ Nucleophilic addition reaction.

90. Ans (4)

MnO_2 oxidised only

1° Allylic and Benzylic Alcohol into Aldehyde.

2° Allylic and Benzylic Alcohol into Ketone.

91. Ans (1)

NCERT-XII, Pg. # 69

92. Ans (2)

NCERT-XII, Pg. # 74-75

93. Ans (4)

NCERT-XII, Pg. # 92

94. Ans (3)

NCERT-XII, Pg. # 61

95. Ans (1)

NCERT Pg. # 86, 87(E), 94, 96(H)

96. Ans (2)

NCERT-XII, Pg. # 54

97. Ans (1)

It is an annual plant with short life span and give results within 2−3 months.

98. Ans (3)

The correct answer is: 3.

5 : 3

In a dihybrid cross, the F_2 generation typically results in a 9:3:3:1 phenotypic ratio. However, if you're specifically looking at the ratio of parental (non-recombinant) to recombinant combinations in the F_2 generation, it generally tends to follow a 5:3 ratio when considering the linked genes.

99. Ans (1)

Correct Answer is (1)

100. Ans (2)

NCERT-XII, Pg. # 74

101. Ans (4)

NCERT-XII, Pg. # 90, 91

102. Ans (3)

NCERT-XII, Pg. # 87

103. Ans (3)

NCERT-XII, Pg. # 88

104. Ans (3)

NCERT-XII, Pg. # 111, 122

105. Ans (1)

NCERT-XII_Pg.# 96, 97

106. Ans (3)

length of DNA = No. of Bp × 3.4 A

= 23 × 3.4 Å

= 78.2 Å

107. Ans (1)

NCERT-XII, Pg. # 97

108. Ans (4)

NCERT-XII, Pg. # 91,98

109. Ans (3)

NCERT-XII, Pg. # 97

110. Ans (4)

NCERT-XII, Pg. # 105

111. Ans (3)

NCERT-XII, Pg. # 96

112. Ans (4)

NCERT-XII, Pg. # 92,93

113. Ans (3)

NCERT-XII, Pg. # 102

114. Ans (2)

Explanation:

Assertion (A) is correct: Many prey species have evolved adaptations such as thorns (in plants), chemical defenses (like alkaloids or toxins), camouflage (like stick insects or leaf-like insects), mimicry, and more to avoid being detected, captured, or eaten by predators. These are classic examples of anti-predatory adaptations discussed in ecology.

Reason (R) is not correct: The statement that "predators are prudent and they overexploit their prey" is contradictory. In ecological terms, predators are generally not prudent in the sense of showing self-restraint; however, natural selection favors predators that do not overexploit their prey, because doing so would lead to extinction of both prey and predator populations. So, the idea of predators overexploiting prey contradicts the observed balance in stable ecosystems.

115. Ans (2)

Resource partitioning is a mechanism where **closely related species** share **limited resources** by occupying **slightly different niches.** This reduces **direct competition**, allowing them to **co-exist in the same habitat.**

For example, different species of **warblers** feeding at different heights of the same tree.

116. **Ans (3)**

Question Explanation:

How many on the given population interections are positive type of interaction ?

Orchids and bees, fig and wasp, barnacles and whale, goat and abingdon tortoise, flamingoes and fish.

Solution:

- Orchids and bees
- Fig and wasp
- Barnacles and whale (commensalism)

Hence, Correct Option: 3

117. **Ans (4)**

NCERT-XII_Pg. # 197

118. **Ans (2)**

Explain Question : The question is asking: "Which pairs of ecological interactions and examples are correctly matched?"

Concept: Population Interaction

Solution :

(A) Co-evolution ⟶ Cuckoo & crow

Although the primary interaction is brood parasitism, over evolutionary time the cuckoo and crow exert selective pressures on each other (cuckoo egg mimicry vs crow egg recognition). This reciprocal evolutionary influence is exactly what co-evolution describes. Correct.

(B) Mutualism ⟶ Fig and wasp

Classic example of obligate mutualism, where both species benefit: the fig gets pollinated, and the wasp gets a site for reproduction. Correct.

(C) Sexual deceit ⟶ Shown by opuntia

Some plants exhibit sexual deceit to attract pollinators without reward, but Opuntia is not the standard example used. Incorrect.

(D) Connell's elegant ⟶ Resource partitioning

Connell's barnacle experiment studied competition, not "resource partitioning." Incorrect .

So, the correct matches are indeed A and B, which is Option 2.

Final answer: option (2)

119. **Ans (4)**

Question Explanation : The question is asking: "Which statement is NOT a characteristic of predation?"

Concept: Predation

Solution:

Predation does help maintain species diversity by controlling dominant prey species and preventing competitive exclusion.

Prudent predator ⟶ True

Acts as conduit for energy ⟶ True

Maintains prey population ⟶ True

Does not maintain species diversity ⟶ False

Final answer: The Correct Answer: 4 Does not maintain species diversity in an ecosystem

120. **Ans (4)**

NCERT Pg. # 197

121. **Ans (1)**

NCERT-XII, Pg. # 261

122. **Ans (3)**

NCERT Pg. # 223, Hindi Pg. # 243

123. **Ans (2)**

NCERT Pg. # 245, 246

124. **Ans (3)**

NCERT-XII, Pg # 264,265

125. **Ans (2)**

NCERT Pg. # 220

126. **Ans (2)**

NCERT Pg. # 220

127. **Ans (3)**

NCERT-XII_Pg. # 222

128. **Ans (2)**

NCERT-XII_Pg. # 222

129. **Ans (1)**

NCERT-XII_Pg. # 211

130. **Ans (3)**

NCERT-XII_Pg. # 213

131. **Ans (4)**
NCERT-XII_Pg. # 210

132. **Ans (4)**
NCERT-XII_Pg. # 213

133. **Ans (3)**
NCERT-XII, Pg. # 212, 213

134. **Ans (4)**
Question Explanation –
The question asks for the factor that does NOT increase the rate of decomposition in an ecosystem.
Concept –
Factors affecting decomposition
Solution –
Decomposition is faster when detritus is rich in sugars, the environment is warm and moist, and aerobic soil microbes are present.
Detritus rich in lignin and chitin is complex and resistant to breakdown, so it slows down decomposition.
Final Answer – Option 4

135. **Ans (3)**
NCERT-XII-Pg. # 219-220

136. **Ans (3)**
Ques Explanation – The question is asking how Ascaris infection usually occurs.
Concept – Mode of transmission of parasitic worms
Solution – Ascaris infection occurs when a person ingests eggs present in contaminated food or water, especially drinking water containing Ascaris eggs.
Final Answer – Option 3

137. **Ans (1)**
Explain Question : sustained high fever of 39°-40°C, weakness, stomach pain, constipation are common symptoms of ?**Concept :** This question is based on Bacterial disease.
Solution : All these symptoms belongs to Typhoid fever.
Final Answer : option (1)

138. **Ans (3)**
Question Explanation: The question is based on the Good Humour (Humoral) hypothesis, which related personality and disease conditions to the dominance of body humours.
Concept: Health
Solution: A person with black bile was believed to have a hot personality associated with fever.
Final answer: Belonged to hot personality having fevers

139. **Ans (4)**
NCERT Pg.#130-131.

140. **Ans (2)**
NCERT Pg.#137.

141. **Ans (3)**
NCERT Pg.#138.

142. **Ans (2)**
NCERT-XII, Pg. # 136, Para-7.2.4

143. **Ans (4)**
Question Explanation: Understanding the type of immunity provided by antivenom.

Concept: vaccination and immunisation

Solution:
Assertion: Injection of antivenom provides passive immunity because it directly supplies antibodies, not by stimulating the body to produce them.
Reason: Antivenom contains preformed antibodies against snake venom = correct
Since the reason correctly explains why the assertion is true, both statements are true and linked.

Final Answer: Option 4

144. Ans (4)

Explaining the Question :- Which statement about drug/alcohol abuse is false ?

Concept:- This question is based on Effects of drug and alcohol abuse

Solution :-

• Statement 4 is false

Withdrawal symptoms can be severe and even life-threatening, especially with substances like alcohol and opioids (e.g., delirium tremens, seizures)

All other statements correctly describe the harmful effects of drug/alcohol abuse.

Final Answer :- 4

145. Ans (2)

NCERT-12th Pg No. # 132, 133

146. Ans (2)

NCERT-XII, Pg. 159

147. Ans (2)

NCERT-XII, Pg. # 147, 148, 149

148. Ans (4)

NCERT Pg.# 142-143

149. Ans (1)

Question Explanation –

The question asks which given combination does not belong to cannabinoids.

Concept –

Cannabinoids and their sources

Solution –

Cannabinoids are obtained from Cannabis plant products such as marijuana, bhang, ganja, charas, and hashish.

Datura contains tropane alkaloids, and LSD is a synthetic hallucinogen, so they are not cannabinoids.

Final Answer – Option 1

150. Ans (4)

Question Explanation : Alfred Wallace's study location.

Concept : Biogeographical distribution.

Solution : Alfred Russel Wallace worked in the **Malay Archipelago**. His extensive fieldwork there led him to independently formulate a theory of evolution similar to Darwin's. He also observed a sharp divide in the fauna, now known as the **Wallace Line**.

Final Answer : Option (4) Malay Archepelago

151. Ans (1)

Question explanation: Identify the correct chronological order of human evolution from early hominids to modern humans.

Concept: Human evolution timeline

Solution:

- The correct evolutionary sequence is:
  - Homo habilis ⟶ first tool-making human-like species
  - Homo erectus ⟶ more advanced, upright posture
  - Neanderthal ⟶ well-developed brain and culture
  - Homo sapiens sapiens ⟶ modern humans
- Other options are incorrect due to wrong sequence or misplaced species (e.g., Australopithecus or Ramapithecus placed incorrectly).

Final answer: Option 1

152. Ans (2)

Question Explanation : Identify the ancestor-A in the family tree of dinosaurs in the given figure

Concept : This question is based on evolution of reptiles and origin of dinosaurs.

Solution : The ancestor-A in the family tree of dinosaurs in the given figure is Thecodonts.

Final Answer : Option (3)

153. **Ans (4)**
NCERT Pg # 124,125

154. **Ans (3)**
NCERT XIIth_Pg. No. 120,121

155. **Ans (3)**
NCERT Pg. # 117

156. **Ans (3)**
NCERT Pg.# 123-124

157. **Ans (1)**
NCERT Pg. # 122

158. **Ans (3)**
NCERT-XII, Pg. # 180

159. **Ans (4)**
NCERT XII, Pg # 179

160. **Ans (1)**
Plasmid vector is the most commonly used vector for introducing nematode-specific genes into a host plant like tobacco.
Plasmids are small, circular DNA molecules that can replicate independently within a bacterial cell. They can be easily manipulated in the laboratory to carry foreign DNA, such as the nematode-specific genes. Once the plasmid containing the desired gene is introduced into a bacterium, it can be transferred to plant cells through methods like Agrobacterium-mediated transformation or direct gene transfer.
While other vectors like cosmids, bacteriophage vectors, and BACs can also be used for gene transfer, plasmids are often preferred due to their ease of manipulation and high efficiency of transformation.

161. **Ans (4)**
NCERT Pg. No. # 179, 180

162. **Ans (1)**
NCERT (XIth) Pg. # 179, 180, 183

163. **Ans (2)**
The milk of the transgenic cow 'Rosie' was nutritionally more balanced for human babies because it contained Human alpha-lactalbumin.

164. **Ans (4)**
NCERT XII, Pg. # 183, 184

165. **Ans (3)**
NCERT Pg. # 184, 185

166. **Ans (2)**
NCERT XII, Page # (E) 211, (H) 232

167. **Ans (4)**
NCERT XII, Pg # 165

168. **Ans (3)**
NCERT XII, Pg # 168

169. **Ans (3)**
NCERT Pg.#171 (E), 187 (H)

170. **Ans (2)**
NCERT 12th, Page No. # 168,167

171. **Ans (4)**
NCERT XIIth Pg.#199

172. **Ans (4)**
NCERT XII, Pg # 171

173. **Ans (2)**
NCERT XII, Pg # 170

174. **Ans (3)**
Question explanation: Asks to arrange the steps of recombinant DNA technology.
Concept: Steps of recombinant DNA technology
Solution: The correct order of steps of RDT are –
(I) Isolation of DNA fragment
(II) Amplification of gene of interest.
(III) Ligation of DNA fragments into vector
(IV) Insertion of r-DNA into tied..
(V) Extraction of desired gene product.
Final Answer: III, II, IV, V, I

175. **Ans (1)**
NCERT-XII, Pg # 166, Fig.9.1

176. **Ans (2)**
NCERT XII, Pg # 171

177. **Ans (4)**
NCERT Pg. # 171,172,174

178. **Ans (1)**
NCERT-XII Pg. # 142

179. **Ans (3)**
NCERT Pg. # 179

180. **Ans (3)**
Explain Question: The question is asking which statement about the naming system of restriction endonuclease enzymes is incorrect.
Concept: Restriction endonuclease enzymes
Solution: Rules for naming restriction enzymes:
First letter ⟶ from genus
Next two letters ⟶ from species
Roman numbers ⟶ show the order of isolation from that bacterial strain, not discovery
Option 3. Incorrect (It says "order of discovery," but it should be "order of isolation")
Final answer: Option 3